
A Median Perspective on Unlabeled Data for Out-of-Distribution Detection

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Abstract

1 Out-of-distribution (OOD) detection underpins robust and reliable machine learning,
2 governing whether models deployed in open-world environments can be trusted
3 to recognize inputs that fall outside their training distribution. A prominent line
4 of recent work exploits unlabeled in-the-wild data to strengthen detection, yet
5 converting such data into a dependable training signal is intrinsically difficult: wild
6 data constitutes an unlabeled mixture of in-distribution (InD) and OOD samples,
7 and the absence of a clean, separable set of OOD examples obstructs the training
8 of an effective detector. We introduce Medix, a median-centric framework that
9 identifies candidate outliers directly from this unlabeled mixture. The choice of
10 the median is principled rather than incidental. As a robust estimator of central
11 tendency, it is by construction insensitive to a substantial fraction of contaminating
12 outliers, which makes an element-wise median statistic a natural filtering criterion
13 for separating InD from OOD gradients. The outliers surfaced by this filter, paired
14 with the labeled InD data, then supervise a binary OOD detector. On the theoretical
15 side, we analyze an idealized median-filtering objective and derive two-sided error
16 bounds that, under coordinate-wise sub-Gaussian gradient assumptions, control the
17 misclassification rate for both inliers and outliers; these bounds articulate precisely
18 the regime in which median filtering is reliable and thereby motivate the design
19 of Medix. Empirically, Medix delivers strong and consistent gains: across eleven
20 InD–OOD benchmark pairs and against twenty competitive baselines spanning
21 both InD-only and wild-data methods, it attains the best FPR95 and AUROC in
22 every reported comparison, with especially pronounced reductions in FPR95.

1 Introduction

24 Deploying machine learning models in real-world applications inevitably exposes them to safety
25 and reliability risks, foremost among them the presence of out-of-distribution (OOD) data. Such
26 inputs originate from categories the model was never trained to recognize, and a trustworthy system
27 must abstain on them rather than commit to a confident in-distribution label. Neural networks are
28 structurally ill-equipped for this demand: absent an explicit detection mechanism, they routinely emit
29 overconfident predictions on OOD inputs at test time (Nguyen et al., 2015), so equipping a deployed
30 model with a dedicated detector is a prerequisite for safe operation rather than an optional refinement.

31 Detecting OOD samples at inference is at once critical and intrinsically hard: because models never
32 observe the unknown distribution during training, they possess no direct signal for separating OOD
33 from in-distribution (InD) data. A productive recent response leverages additional “in-the-wild” data
34 to sharpen detection (Katz-Samuels et al., 2022a; Du et al., 2024a). In particular, Katz-Samuels
35 et al. (2022a) regularize training with unlabeled wild data while retaining the primary objective
36 of classifying labeled InD inputs. The appeal of wild data is at once practical and statistical: it is
37 abundant and essentially free to harvest once a model is deployed in its operating environment, and,

38 crucially, it exposes the model to the very distribution of OOD inputs it will encounter at test time,
39 yielding markedly more robust detection than InD data alone can support.

40 Exploiting wild data, however, is far from straightforward, precisely because it arrives as an unlabeled
41 mixture of InD and OOD samples. Lacking a clean, separable set of OOD examples, a learner
42 observes only samples drawn from this mixed distribution and receives no indication of whether any
43 given point is InD or OOD—an absence of supervision that substantially complicates the design
44 of a reliable detection rule. The setting therefore remains underexplored and rich with opportunity.
45 Compounding the methodological gap, the literature offers little formal theory: to the best of our
46 knowledge, Du et al. (2024a) is the sole prior work to supply a rigorous foundation for the “in-the-wild”
47 setting.

48 In parallel, median-based estimators have proven remarkably effective in robust data pruning (Acharya
49 et al., 2024), where their tolerance to corrupted samples is decisive. This convergence—an unlabeled
50 mixed-data problem that demands robustness, and a robust statistic engineered precisely for
51 contaminated data—motivates the central question of this paper:

How can median-based methods leverage unlabeled wild data to facilitate OOD
detection with theoretical guarantees?

52 We answer this question affirmatively by developing a median-centric perspective on OOD detection.
53 Concretely, we cast outlier identification as a median-based subset-selection problem and instantiate
54 a tractable greedy solver in Medix, an algorithm that extracts OOD samples from wild data at a low
55 empirical error rate. The framework is grounded in a theoretical analysis of the idealized median-
56 filtering objective and validated by experiments that demonstrate both its accuracy and its efficiency,
57 placing our work among the few to offer such an analysis for the unlabeled “in-the-wild” setting. The
58 core theoretical result characterizes when median-based filtering separates inliers from outliers in an
59 unlabeled mixture, bounding the fraction of InD samples erroneously flagged as outliers. This fraction
60 is governed by two complementary effects: a *contamination effect*, which captures the perturbation
61 induced by OOD points and remains controlled provided the OOD proportion stays below one half;
62 and a *concentration effect*, which exploits the coordinate-wise sub-Gaussian behavior of InD gradients
63 to bound their deviation from the mean gradient. Jointly, and under these assumptions, the two effects
64 imply that with high probability only a small fraction of InD points are misclassified—clarifying the
65 regime in which the median criterion is trustworthy and directly motivating the filtering rule at the
66 heart of Medix.

67 We evaluate Medix against two families of methods: (1) those trained solely on InD data, and (2)
68 those trained jointly on InD data and an auxiliary unlabeled dataset. On CIFAR-100 (Krizhevsky
69 et al., 2009), Medix improves markedly over the strong contrastive baseline KNN+ (Sun et al., 2022),
70 reducing FPR95 by an average of 40.98%. Crucially, methods such as Outlier Exposure (Hendrycks
71 et al., 2019) presuppose a clean auxiliary dataset—imposing the strong distributional assumption
72 that the auxiliary data is fully separable from the InD data—whereas Medix attains superior results
73 without any such assumption, and therefore applies directly to genuinely mixed wild data. Against
74 WOODS (Katz-Samuels et al., 2022a), the strongest wild-data baseline, Medix lowers the average
75 FPR95 by 1.32% on CIFAR-100 and 2.60% on CIFAR-10, and does so while dispensing with the
76 batch-level mixing structure that WOODS requires.

77 Our contributions are threefold:

78 **C1) A median-centric framework for OOD detection.** We propose Medix, which first filters
79 outliers from unlabeled wild data through a median-centric greedy procedure and then trains
80 an OOD detector on the extracted outliers together with the labeled InD samples. The
81 filtering stage is our central methodological contribution: it operates on genuinely mixed
82 wild data at the dataset level, without assuming access to a clean OOD set and without the
83 batch-level mixing structure required by prior wild-data methods.

84 **C2) Two-sided theoretical analysis of the median-filtering objective.** We analyze the idealized
85 subset-selection problem that Medix approximates and derive two-sided bounds on the
86 misclassification rate for InD samples flagged as outliers and, symmetrically, for OOD
87 samples retained as inliers. Under coordinate-wise sub-Gaussian gradient assumptions, both
88 rates decompose into an interpretable contamination effect and a concentration effect, and

the analysis makes explicit the regime—chiefly, an OOD proportion below one half for the inlier bound—in which median filtering is provably reliable.

C3) Comprehensive empirical validation. We evaluate Medix across eleven InD–OOD pairs against 20 competitive baselines spanning InD-only and wild-data methods, and find that it attains the best FPR95 and AUROC in every reported comparison. A controlled synthetic study, in which the ground-truth OOD labels are known, further shows that Medix extracts outliers at a low error rate—as low as 12.5% (see Figure 2)—instantiating the low-error behavior predicted by our analysis.

2 Preliminaries and Problem Setup

We first frame the OOD detection problem and then formally specify the data setup, model architecture, loss functions, and learning objective on which the remainder of the paper builds.

Labeled In-Distribution Data. Let $\mathcal{X}$ denote the input space and $\mathcal{Y} = \{1, \dots, K\}$ the label space, which together fix the structure of the InD data. The labeled dataset $\mathcal{S}_{\text{in}} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$ comprises n pairs drawn independently and identically distributed (i.i.d.) from $\mathbb{P}_{XY}$, an unknown joint distribution over $\mathcal{X} \times \mathcal{Y}$. We write $\mathbb{P}_{\text{in}}$ for the marginal of $\mathbb{P}_{XY}$ on $\mathcal{X}$ —the underlying distribution of InD inputs—and use $\mathcal{S}_{\text{in}}$ to train the InD model.

Out-of-distribution detection. We consider the practical deployment scenario in which a model trained on labeled InD data is subsequently exposed to inputs from classes absent from training, i.e., inputs with label $y \notin \mathcal{Y}$. On such OOD inputs the model must abstain rather than commit to an InD label. The inference-time task is therefore a binary decision: to determine whether a given input is drawn from the InD distribution or originates from an OOD source.

Unlabeled wild data. A primary obstacle in OOD detection is the scarcity of labeled OOD samples: the space of possible OOD inputs is effectively unbounded, rendering the collection of labeled examples both costly and impractical. Following Katz-Samuels et al. (2022a), we instead incorporate unlabeled wild data, $\mathcal{S}_{\text{wild}} = \{\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_m\}$, into the learning framework, thereby reflecting realistic deployment conditions. Wild data is a blend of InD and OOD samples and is readily harvested once a model pre-trained on $\mathcal{S}_{\text{in}}$ is deployed. Consistent with Du et al. (2024a); Katz-Samuels et al. (2022a), we adopt the Huber contamination model (Huber, 1964) to characterize the marginal distribution of the wild data:

$$\mathbb{P}_{\text{wild}} := (1 - \pi)\mathbb{P}_{\text{in}} + \pi\mathbb{P}_{\text{out}}, \quad \pi \in (0, 1], \quad (1)$$

where π is the contamination proportion and $\mathbb{P}_{\text{out}}$ denotes the OOD distribution over $\mathcal{X}$. The degenerate case $\pi = 0$ corresponds to wild data devoid of OOD samples and renders the problem trivial; we therefore restrict attention to the informative regime $\pi \in (0, 1]$.

Models and Loss Functions. Let $f_\phi : \mathcal{X} \rightarrow \mathbb{R}^K$ be the InD classifier parameterized by $\phi \in \Phi$, whose output is a soft probability distribution over the $K = |\mathcal{Y}|$ InD classes, and let $\ell : \mathbb{R}^K \times \mathcal{Y} \rightarrow \mathbb{R}$ denote the associated supervised loss on labeled InD data. For detection proper, we introduce a dedicated binary classifier $g_\theta : \mathcal{X} \rightarrow \mathbb{R}$ parameterized by $\theta \in \Theta$ and trained under the loss $\ell_b(g_\theta(x), y_b)$ with $y_b \in \mathcal{Y}_b := \{y_+, y_-\}$, where $y_+ > 0$ encodes the InD class and $y_- < 0$ the OOD class, so that the sign of g_θ realizes the detection decision.

Learning objective. Our framework jointly trains the OOD detector g_θ and the multi-class classifier f_ϕ , drawing on both the InD data $\mathbb{P}_{\text{in}}$ and the wild data $\mathbb{P}_{\text{wild}}$. At test time we quantify performance through three complementary metrics:

$$\begin{aligned} \downarrow \text{FPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{out}}^{\text{test}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{TPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{in}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{Acc}(f_\phi) &= \mathbb{E}_{(\mathbf{x}, y) \sim \mathbb{P}_{XY}} [\mathbb{I}\{f_\phi(\mathbf{x}) = y\}], \end{aligned}$$

where $\mathbb{I}\{\cdot\}$ is the indicator function and $\mathbb{P}_{\text{out}}^{\text{test}}$ denotes the OOD test distribution. A detector is effective precisely when it simultaneously drives the false positive rate down while holding both the true positive rate and InD classification accuracy high.

3 Method: Median-Centric Framework for OOD Detection

We now present **Medix**, a learning paradigm for OOD detection that harnesses unlabeled wild data. By moving beyond methods that rely exclusively on InD data, Medix is expressly suited to open-world environments in which models are routinely confronted with previously unseen inputs. The framework decomposes into two tightly coupled stages: **(1) Outlier Extraction**, a gradient-space filtering procedure that isolates candidate OOD samples from the unlabeled wild data (Section 3.1); and **(2) Detector Training**, in which a binary OOD detector is trained on the labeled InD data together with the outlier candidates surfaced in stage 1 (Section 3.2), following the protocol of Du et al. (2024a). As the following sections establish, this two-stage design extracts OOD data from the unlabeled wild mixture effectively and, in turn, yields a dependable detector for deployment in dynamic, open-world settings.

3.1 Extracting Candidate Outliers from the Wild Data

To isolate potential outliers from the wild mixture $\mathcal{S}_{\text{wild}}$, Medix adopts an optimization-based formulation grounded in the parameter gradients of a classification model f_ϕ trained exclusively on the InD dataset $\mathcal{S}_{\text{in}}$. The rationale is that the loss gradient of a sample, taken with respect to the model parameters, encodes how far that sample departs from InD behavior, and therefore supplies a discriminative signal for filtering. We formalize the procedure below.

Reference gradient estimation from InD data. The first step estimates a reference gradient from the InD dataset $\mathcal{S}_{\text{in}}$. We train the classifier f_ϕ on $\mathcal{S}_{\text{in}}$ by empirical risk minimization (ERM),

$$\phi_{\mathcal{S}_{\text{in}}} \in \arg \min_{\phi \in \Phi} \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi), \text{ where } \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi) = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \ell(f_\phi(\mathbf{x}_i), y_i), \quad (2)$$

where $\phi_{\mathcal{S}_{\text{in}}}$ denotes the learned parameters. With the classifier held fixed, we then form the mean gradient $\bar{\nabla}_{\text{in}}$ by averaging the parameter gradients of the loss over the InD data:

$$\bar{\nabla}_{\text{in}} = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}}(\mathbf{x}_i), y_i). \quad (3)$$

This quantity $\bar{\nabla}_{\text{in}}$ serves as the reference gradient, against which the deviation of any candidate point is measured.

Motivation. Our filtering rule rests on a simple, testable hypothesis: as the number of OOD samples in the wild dataset $\mathcal{S}_{\text{wild}}$ grows, the element-wise median of the wild gradients drifts progressively further from the mean InD gradient $\bar{\nabla}_{\text{in}}$. We verify this hypothesis directly in a controlled experiment with CIFAR-10 (Krizhevsky et al., 2009) as the InD dataset and SVHN (Netzer et al., 2011) as the OOD dataset. Here $\mathcal{S}_{\text{wild}}$ is initialized with 10,000 CIFAR-10 samples held disjoint from the data used to train $\phi_{\mathcal{S}_{\text{in}}}$ and to compute $\bar{\nabla}_{\text{in}}$. We then inject SVHN OOD samples into $\mathcal{S}_{\text{wild}}$ incrementally and track the L_2 -norm deviation between $\bar{\nabla}_{\text{in}}$ and the element-wise median (EWM) of the wild gradients,

$$\left\| \bar{\nabla}_{\text{in}} - \text{EWM} \left(\left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}_{\text{wild}}} \right) \right\|.$$

As Figure 1 shows, the L_2 -norm deviation increases clearly and monotonically with the injected OOD fraction, confirming the hypothesis. This monotonic relationship is more than a qualitative observation: it directly furnishes the stopping criterion for our algorithm, which halts once the change in L_2 -norm deviation between consecutive iterations falls below a threshold ϵ . We detail this mechanism in the next section.

Filtering potential outliers from unlabeled wild data. Building on the monotonic trend of Figure 1, we cast outlier identification as the following subset-selection problem, whose solution yields the outlier subset $\mathcal{S}_{\text{out}}^*$ within $\mathcal{S}_{\text{wild}}$:

$$\mathcal{S}_{\text{in}}^* = \arg \min_{\mathcal{S} \subseteq \mathcal{S}_{\text{wild}}} \left\| \bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}}) \right\|, \text{ where } G_{\mathcal{S}} = \left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}}. \quad (4)$$

Here $\text{EWM}(\cdot)$ is the element-wise (coordinate-wise) median and $\hat{y}_{\tilde{\mathbf{x}}_i}$ is the top-1 pseudo-label predicted by the InD classifier $f_{\phi_{\mathcal{S}_{\text{in}}}}$ for the wild sample $\tilde{\mathbf{x}}_i$ —the argmax of its softmax output—which supplies the target needed to evaluate the loss gradient on an unlabeled point; for brevity we write $G_{\mathcal{S}} = \{\nabla \ell(\tilde{\mathbf{x}}_i)\}_{i \in \mathcal{S}}$.

Intuitively, (4) seeks the subset $\mathcal{S} \subseteq \mathcal{S}_{\text{wild}}$ whose gradient EWM lies closest to the reference $\bar{\nabla}_{\text{in}}$. By the monotonic relationship established in Figure 1, this minimizing subset $\mathcal{S}_{\text{in}}^*$ is precisely the one whose gradients are least contaminated and therefore best represents the InD data within $\mathcal{S}_{\text{wild}}$; its complement $\mathcal{S}_{\text{out}}^* = \mathcal{S}_{\text{wild}} \setminus \mathcal{S}_{\text{in}}^*$ then recovers the OOD data. The theoretical analysis of Section 4 concerns exactly this idealized minimizer.

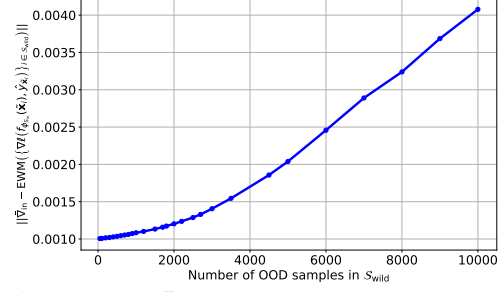


Figure 1: The L_2 -norm deviation between the InD reference gradient $\bar{\nabla}_{\text{in}}$ and the EWM of the wild gradients grows monotonically as SVHN OOD samples are incrementally injected into $\mathcal{S}_{\text{wild}}$ (CIFAR-10 InD, SVHN OOD). This monotonic trend underpins both the objective in (4) and the stopping criterion of Algorithm 1.

Solving (4) exactly is combinatorial and becomes computationally prohibitive as $|\mathcal{S}_{\text{wild}}|$ grows, since it ranges over all subsets of the wild set. We therefore develop a tractable greedy approximation based on a leave-one-out criterion, summarized in Algorithm 1; the algorithm is a heuristic that targets the objective in (4) rather than an exact solver. It proceeds iteratively, ranking wild samples by the gradient deviation each induces relative to the InD reference and removing the most anomalous at every step.

At each iteration, the algorithm computes the EWM of the gradients of the samples remaining in the wild set $\mathcal{S}$ and measures its L_2 distance to the InD reference, $d_t = \|\text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}}\|$. It then quantifies, for every candidate $i \in \mathcal{S}$, the reduction in this distance that would follow from deleting i —forming $\text{EWM}(G_{\mathcal{S} \setminus \{i\}})$ and the corresponding drop $\delta_i = d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$. Because a genuine OOD point pulls the median away from $\bar{\nabla}_{\text{in}}$, the samples whose removal most sharply decreases this distance are exactly those most likely to be outliers; the algorithm accordingly flags the k samples with the largest δ_i as OOD and removes them. The procedure repeats until no remaining sample yields a significant drop, or a maximum iteration count T is reached. The convergence criterion—terminating once the change in L_2 distance between successive iterations falls below the threshold ϵ —follows directly from the monotonic trend documented in Figure 1: once the bulk of the OOD mass has been excised, the distance flattens to a small residual value, and halting at that point prevents the algorithm from erroneously discarding InD samples and thereby eroding detection performance. This early-stopping behavior is what keeps the greedy heuristic aligned, in practice, with the low-error regime characterized by our analysis.

Algorithm 1 Iterative Outlier Detection via Medix

Require: $\bar{\nabla}_{\text{in}}, G_i = \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i}})$, maximum iterations T , hyperparameters ϵ, k

- 1: Initialize $\mathcal{S} \leftarrow \mathcal{S}_{\text{wild}}$ (wild set), $\mathcal{O} \leftarrow \emptyset$ (outliers), $t \leftarrow 0$ (iteration), $d_t \leftarrow 0$, $\delta_{\text{max}} \leftarrow \infty$ (deviations), $\mathcal{I}_k \leftarrow \emptyset$
- 2: **while** $t \leq T$ or $|\delta_{\text{max}}| > \epsilon$ **do**
- 3: $\mathcal{O} \leftarrow \mathcal{O} \cup \{\tilde{\mathbf{x}}_i : i \in \mathcal{I}_k\}$ ▷ Add outliers to set $\mathcal{O}$
- 4: $d_t \leftarrow \|\text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}}\|$ ▷ Compute L2 deviation
- 5: **for each** $i \in \mathcal{S}$ **do**
- 6: $\delta_i \leftarrow d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$
- 7: **end for**
- 8: $\mathcal{I}_k \leftarrow \text{indices}(\text{top-}k(\{\delta_i\}_{i \in \mathcal{S}}))$ ▷ Select top- k indices
- 9: $\mathcal{S} \leftarrow \mathcal{S} \setminus \mathcal{I}_k$ ▷ Remove outliers from $\mathcal{S}$
- 10: $\delta_{\text{max}} \leftarrow \max_{i \in \mathcal{S}} \{\delta_i\}$, $t \leftarrow t + 1$
- 11: **end while**
- 12: **return** $\mathcal{S}_{\text{out}} = \mathcal{O}$ ▷ Return detected outliers

3.2 Training the OOD Detector with Candidate Outliers

With the candidate outlier set $\hat{\mathcal{S}}_{\text{out}}$ extracted from the wild data, we train the OOD detector g_{θ} to sharpen the separation between InD and outlier data, following the protocol of Du et al. (2024a). The objective enforces separability at a decision boundary thresholded at 0, driving g_{θ} toward positive outputs on labeled InD samples $\mathbf{x} \in \mathcal{S}_{\text{in}}$ and negative outputs on the candidate outliers $\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}$

surfaced in stage 1. Concretely, the loss is

$$\begin{aligned}\mathcal{L}_{\mathcal{S}_{\text{in}}, \mathcal{S}_{\text{out}}}(g_\theta) &= \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_\theta) + \mathcal{L}_{\mathcal{S}_{\text{out}}}^-(g_\theta), \\ \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_\theta) &= \mathbb{E}_{\mathbf{x} \in \mathcal{S}_{\text{in}}} \mathbb{I}\{g_\theta(\mathbf{x}) \leq 0\}, \\ \mathcal{L}_{\mathcal{S}_{\text{out}}}^-(g_\theta) &= \mathbb{E}_{\tilde{\mathbf{x}} \in \mathcal{S}_{\text{out}}} \mathbb{I}\{g_\theta(\tilde{\mathbf{x}}) > 0\}.\end{aligned}\tag{5}$$

Because the 0/1 loss in (5) is non-differentiable, we optimize a smooth sigmoid-based surrogate that remains aligned with the original objective while being tractable under gradient-based training. The framework simultaneously retains the InD risk of Equation 2, which safeguards predictive accuracy on InD inputs, so that detection is sharpened without sacrificing classification. This joint objective strengthens the generalization of g_θ and yields robust detection of OOD samples drawn from $\mathbb{P}_{\text{out}}$, as our experiments confirm.

4 Theoretical Analysis

We now develop the theoretical analysis that motivates Medix’s filtering stage. The two theorems below concern the idealized subset-minimization objective of (4)—the exact minimizer that Algorithm 1 approximates—and furnish upper bounds on the misclassification rates for InD and OOD points, respectively. Read together, they give a two-sided account of when EWM filtering succeeds: they identify the conditions under which neither type of error can be large, and, equally informatively, the regimes in which each bound ceases to be tight. The results should thus be understood as characterizing the median-filtering principle rather than as end-to-end guarantees for the greedy implementation. Complete proofs are deferred to Appendix C.

Theorem 4.1 (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$, and fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the EWM filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

Theorem 4.1 decomposes the inlier misclassification rate into two interpretable effects: a *concentration term*, which captures statistical fluctuations in the InD gradients and vanishes as the InD sample size grows, and a *contamination term*, which quantifies how OOD points perturb the EWM. The practically important consequence is that the bound remains nontrivial for any contamination ratio $\pi < 0.5$ —that is, whenever the InD samples form the majority of the wild mixture, the median criterion is provably robust to a substantial OOD presence, and the contamination term $\pi/[2(1-\pi)]$ stays below one. This addresses one side of the problem; we now turn to its complement: *how many true outliers evade the filter and are mistakenly retained?* The next theorem bounds this OOD misclassification rate.

Theorem 4.2 (Outlier Misclassification Bound). *Assume that the gradients of OOD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ_{out}^2 . Suppose the mean OOD gradient μ_{out} satisfies a separation condition:*

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d},$$

for some $\Delta > 0$. Then, for any tolerance $\epsilon \in (0, \Delta)$ and confidence level $\delta \in (0, 1)$, with probability at least $1 - \delta$, the outlier misclassification rate satisfies:

$$\text{ERR}_{\text{out}} \leq \underbrace{2d \exp\left(-\frac{(\Delta - \epsilon)^2}{2\sigma_{\text{out}}^2}\right)}_{\text{Separation term}} + \underbrace{\sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}}_{\text{Concentration term}} + \underbrace{\frac{1-\pi}{2\pi}}_{\text{Contamination term}}.$$

Theorem 4.2 complements the inlier bound by controlling the fraction of OOD points mistakenly retained as InD. Taken together, Theorems 4.1 and 4.2 give a two-sided account of median filtering: the inlier bound is informative when InD points are the majority ($\pi < 0.5$), and the outlier

bound is informative in the symmetric regime, when OOD points are not a small minority. Both misclassification rates are governed by the interplay of three effects, which we now examine in turn.

Contamination Effect. In Theorem 4.1, the median criterion remains stable even when a substantial fraction π of the wild data is OOD, provided $\pi < 0.5$. This is encoded in the $\pi/[2(1 - \pi)]$ term, which grows with π and measures the risk that OOD gradients skew the EWM. Theorem 4.2 carries the mirror-image penalty $\frac{1-\pi}{2\pi}$, which quantifies the difficulty of isolating OOD points when they are underrepresented; we note candidly that this term exceeds one—and the outlier bound consequently becomes vacuous—once π falls below $1/3$, so the outlier guarantee is meaningful only when OOD samples are not a small minority. The two terms thus articulate a single, coherent principle, together with its limits: median filtering is provably reliable for a given class precisely when that class is not the minority, and neither bound claims control in the opposite regime.

Concentration Effect. Both bounds exploit the coordinate-wise sub-Gaussian behavior of the gradients. For InD data, sub-Gaussianity forces most gradients to concentrate near $\bar{\nabla}_{\text{in}}$, keeping the median stable even at finite sample sizes; for OOD data, concentration around μ_{out} bounds the probability that a misaligned OOD sample slips past the filter. In both cases the effect decays as $1/\sqrt{m}$ in the per-class sample count m , so larger wild sets tighten the bounds.

Separation Effect. Unique to Theorem 4.2, this effect specifies how far the OOD mean gradient must lie from the InD mean gradient for OOD samples to be reliably rejected. The exponential term $\exp(-(\Delta - \varepsilon)^2/2\sigma_{\text{out}}^2)$ makes the trade-off explicit: the greater the separation between the two gradient distributions, the exponentially smaller the chance that an OOD gradient evades the filter.

Theorems 4.1 and 4.2 together show that, with high probability, the idealized median filter separates InD and OOD samples in an unlabeled mixture within a well-characterized regime. The fraction of InD samples misclassified as outliers is controlled by contamination and concentration effects, while the fraction of OOD points incorrectly retained is governed by an exponential separation term, a concentration bound, and a reverse contamination effect. These results provide a two-sided characterization of the median-filtering principle underlying Medix, delimiting both when each error type is provably small and the assumptions—coordinate-wise sub-Gaussian gradients and, for the outlier side, sufficient mean separation—under which the guarantees hold.

Remark 4.3. Sub-Gaussian Assumption for Gradient Coordinates. The coordinate-wise sub-Gaussianity of InD gradients is the analytical lever behind our concentration bounds, and it is supported empirically on the datasets we study. Figure 4a shows that the histogram of InD gradient values is bell-shaped and tightly concentrated about its mean, exhibiting the light tails—exponentially decaying, with extreme values rare—characteristic of sub-Gaussian random variables. Figure 4b reinforces this reading: the Q-Q plot (Wilk and Gnanadesikan, 1968) of the empirical gradient quantiles against theoretical Gaussian quantiles tracks the 45-degree reference line closely, indicating near-Gaussian behavior, and since Gaussian variables are a special case of sub-Gaussian ones, this evidence substantiates the assumption for these gradients. *Importantly, the assumption is not indispensable: we further derive a relaxed version of the bounds that dispenses with sub-Gaussianity and holds under merely bounded fourth moments (Theorem C.3, Appendix C.3). The dimensional rate degrades in this broader regime—from an exponential to an $O(1/d)$ decay—yet the finite-sample robustness term is preserved, showing that the qualitative reliability of median filtering does not hinge on strong tail assumptions, even as the quantitative bound loosens.*

5 Experiments

We now demonstrate the efficacy of Medix across a diverse range of InD–OOD dataset pairs, benchmarking it against 20 widely used baselines that span both InD-only and wild-data detectors. All experiments run on NVIDIA A100-SXM4-80GB GPUs, and the code required to reproduce our results will be released upon acceptance.

5.1 Models, Datasets, and Baselines

Datasets. We adopt the experimental protocol of Katz-Samuels et al. (2022a), who introduced the problem of learning OOD detectors from wild data; this choice ensures a fair, like-for-like comparison

against prior work. CIFAR-10 and CIFAR-100 serve as InD datasets ($\mathbb{P}_{\text{in}}$). For OOD testing we employ a suite of natural-image datasets—PLACES365 (Zhou et al., 2017), SVHN, TEXTURES (Cimpoi et al., 2014), LSUN-RESIZE, and LSUN-C (Yu et al., 2015)—as $\mathbb{P}_{\text{out}}$. To construct wild data ($\mathbb{P}_{\text{wild}}$), we combine a subset of the InD data ($\mathbb{P}_{\text{in}}$) with OOD data ($\mathbb{P}_{\text{out}}$) at the default contamination ratio $\pi = 0.5$; for instance, with PLACES365 as the OOD test set, the wild mixture blends CIFAR with PLACES365 and evaluation is likewise performed on PLACES365. We repeat this procedure across every OOD dataset and baseline. The InD CIFAR data is partitioned into two disjoint halves: the first 25,000 images train $\phi_{\mathcal{S}_{\text{in}}}$, and the remaining 25,000 form the InD component of the wild mixture $\mathcal{S}_{\text{wild}}$. All gradients are computed on the penultimate-layer weights, which prior work has shown to be especially informative for OOD detection (Huang et al., 2021a), and the wild data are mixed at the dataset level rather than by batch.

Evaluation metrics. We report three standard metrics: (1) the False Positive Rate of OOD samples at a 95% True Positive Rate on InD samples (FPR95 $\downarrow$), (2) the Area Under the Receiver Operating Characteristic Curve (AUROC $\uparrow$), and (3) InD Classification Accuracy (InD Acc $\uparrow$). The first two quantify detection quality, whereas the third verifies that improved detection is not purchased at the expense of InD performance.

Baselines. Our evaluation spans a broad and competitive set of baselines, organized by whether they train on InD data alone or on both InD and wild data. Among methods trained exclusively on InD data ($\mathbb{P}_{\text{in}}$), we compare against a comprehensive slate of established detectors—Maximum Softmax Probability (MSP) (Hendrycks and Gimpel, 2016), ODIN (Liang et al., 2018), Mahalanobis Distance (Lee et al., 2018a), Energy Score (Liu et al., 2020b), ReAct (Sun et al., 2021), DICE (Sun and Li, 2022), KNN Distance (Sun et al., 2022), and ASH (Djurisic et al., 2022)—all built on a softmax cross-entropy backbone, and we additionally include the contrastive-loss methods CSI (Tack et al., 2020a) and KNN+ (Sun et al., 2022). Among methods that exploit both InD and wild data, we compare against Outlier Exposure (OE) (Hendrycks et al., 2019) and energy-regularization learning (Liu et al., 2020b), which regularize training by suppressing confidence or raising energy on outlier data, together with WOODS (Katz-Samuels et al., 2022a), which introduced the wild-unlabeled-data setting and addresses it via constrained optimization. We further incorporate the recent CONJ (Peng et al., 2024) and DRL (Zhang et al., 2024) baselines to keep the comparison thorough and current. For fairness, every wild-data baseline is trained on the identical unlabeled wild dataset supplied to Medix.

5.2 Experimental Setup

Following WOODS (Katz-Samuels et al., 2022a), the InD classifier $\phi_{\mathcal{S}_{\text{in}}}$ is a Wide ResNet (Zagoruyko, 2016) with 40 layers and a width factor of 2, trained by stochastic gradient descent with momentum 0.9, weight decay 0.0005, and initial learning rate 0.1, for 100 epochs under cosine learning-rate decay, batch size 128, and dropout rate 0.3. The two Medix filtering hyperparameters—the stopping threshold ϵ and the per-iteration removal count k —are selected from $\{5e-5, 5e-4, 5e-3, 5e-2\}$ and $\{4k, 7k, 10k, 20k\}$, respectively, with k scaled to the size of the wild set and the pair chosen to maximize OOD performance; the maximum iteration count T serves only as a safeguard, since termination is governed in practice by the ϵ criterion. As the sensitivity analysis of Appendix A.2 confirms, detection performance is largely insensitive to these choices, so no fine-grained tuning is required. The OOD classifier g_θ is initialized from the pre-trained $\phi_{\mathcal{S}_{\text{in}}}$ and augmented with a linear layer that performs binary classification on the penultimate-layer features. It is fine-tuned for 100 epochs at learning rate 0.001 under the objective of Equation (5), which combines the binary classification loss with the InD classification loss at a relative weight of 10 on the binary term; all remaining training parameters match those used for $\phi_{\mathcal{S}_{\text{in}}}$.

5.3 Results

Our main CIFAR-100 results appear in Table 2, where Medix outperforms every OOD detection baseline on both detection metrics. Two observations stand out. First, methods that exploit both InD and wild data decisively outperform those trained on InD data alone: relative to the best InD-only baseline KNN+, Medix cuts FPR95 by 52.31 points on PLACES365 and 38.24 points on TEXTURES, quantifying the value of in-the-wild data for regularization. Second, and more tellingly, Medix also surpasses the strongest wild-data methods ($\mathbb{P}_{\text{wild}}$): it attains an average FPR95 of 5.42% on CIFAR-

Table 1: OOD detection performance of Medix versus baselines with CIFAR-10 as InD data, across five OOD test sets (SVHN, PLACES365, LSUN-C, LSUN-RESIZE, TEXTURES) and their average. Methods are grouped by whether they use $\mathbb{P}_{\text{in}}$ only or both $\mathbb{P}_{\text{in}}$ and $\mathbb{P}_{\text{wild}}$; lower FPR95 and higher AUROC are better, and every wild-data method sees the same unlabeled wild set. Results are averaged over five runs (reported with standard deviations), with the best entry in each column in **bold**. Medix attains the best FPR95 and AUROC on every OOD set and the best averages (0.80 FPR95, 99.74 AUROC), cutting the average FPR95 of the strongest wild-data baseline WOODS (3.40) by more than fourfold while preserving competitive InD accuracy.

Methods	OOD Datasets												InD ACC $\uparrow$
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average		
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	
Using $\mathbb{P}_{\text{in}}$ only													
MSP	48.49	91.89	59.48	88.20	30.80	95.65	52.15	91.37	59.28	88.50	50.04	91.12	94.84
ODIN	33.35	91.96	57.40	84.49	15.52	97.04	26.62	94.57	49.12	84.97	36.40	90.61	94.84
Mahalanobis	12.89	97.62	68.57	84.61	39.22	94.15	42.62	93.23	15.00	97.33	35.66	93.34	94.84
Energy	35.59	90.96	40.14	89.89	8.26	98.35	27.58	94.24	52.79	85.22	32.87	91.73	94.84
KNN	24.53	95.96	25.29	95.69	25.55	95.26	27.57	94.71	50.90	89.14	30.77	94.15	94.84
ReAct	40.76	89.57	41.44	90.44	14.38	97.21	33.63	93.58	53.63	86.59	36.77	91.48	94.84
DICE	35.44	89.65	46.83	86.69	6.32	98.68	28.93	93.56	53.62	82.20	34.23	90.16	94.84
ASH	6.51	98.65	48.45	88.34	0.90	99.73	4.96	98.92	24.34	95.09	17.03	96.15	94.84
CSI	17.30	97.40	34.95	93.64	1.95	99.55	12.15	98.01	20.45	95.93	17.36	96.91	94.17
KNN+	2.99	99.41	24.69	94.84	2.95	99.39	11.22	97.98	9.65	98.37	10.30	97.99	93.19
Using $\mathbb{P}_{\text{in}}$ and $\mathbb{P}_{\text{wild}}$													
OE	1.13	99.53	19.48	94.88	1.91	98.16	0.54	98.84	7.75	98.56	6.16	97.99	94.12
Energy (w/ OE)	5.24	98.72	14.66	96.18	2.35	99.30	4.85	98.62	10.51	97.10	7.52	97.98	94.24
WOODS	0.17	99.91	10.19	98.05	0.31	99.14	0.11	99.38	6.21	98.13	3.40	98.92	94.74
Medix	0.06	99.98	2.98	99.10	0.01	99.98	0.01	99.98	0.96	99.66	0.80	99.74	93.58
(Ours)	± 0.01	± 0.01	± 0.29	± 0.09	± 0.01	± 0.00	± 0.01	± 0.01	± 0.13	± 0.06	± 0.09	± 0.03	± 0.64

Table 2: OOD detection performance of Medix versus baselines with CIFAR-100 as InD data, across five OOD test sets and their average. Methods are grouped by whether they use $\mathbb{P}_{\text{in}}$ only or both $\mathbb{P}_{\text{in}}$ and $\mathbb{P}_{\text{wild}}$; lower FPR95 and higher AUROC are better, and every wild-data method sees the same unlabeled wild set. Results are averaged over five runs (reported with standard deviations), with the best entry in each column in **bold**. Medix records the best FPR95 and AUROC on every OOD set and the best averages (5.42 FPR95, 98.96 AUROC), improving the average FPR95 over the strongest wild-data baseline WOODS (6.74) by 1.32 points absolute; the accuracy gap to InD-only baselines reflects the halved labeled training set (25,000 vs. 50,000 images), not a shortfall of the method.

Methods	OOD Datasets												InD ACC $\uparrow$	
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average			
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$		
	Using $\mathbb{P}_{\text{in}}$ only													
MSP	84.59	71.44	82.84	73.78	66.54	83.79	82.42	75.38	83.29	73.34	79.94	75.55	75.96	
ODIN	84.66	67.26	87.88	71.63	55.55	87.73	71.96	81.82	79.27	73.45	75.86	76.38	75.96	
Mahalanobis	57.52	86.01	88.83	67.87	91.18	69.69	21.23	96.00	39.39	90.57	59.63	82.03	75.96	
Energy	85.82	73.99	80.56	75.44	35.32	93.53	79.47	79.23	79.41	76.28	72.12	79.69	75.96	
KNN	66.38	83.76	79.17	71.91	70.96	83.71	77.83	78.85	88.00	67.19	76.47	77.08	75.96	
ReAct	74.33	88.04	81.33	74.32	39.30	91.19	79.86	73.69	67.38	82.80	68.44	82.01	75.96	
DICE	88.35	72.58	81.61	75.07	26.77	94.74	80.21	78.50	76.29	76.07	70.65	79.39	75.96	
ASH	21.36	94.28	68.37	71.22	15.27	95.65	68.18	85.42	40.87	92.29	42.81	87.77	75.96	
CSI	64.70	84.97	82.25	73.63	38.10	92.52	91.55	63.42	74.70	92.66	70.26	81.44	69.90	
KNN+	32.21	93.74	68.30	75.31	40.37	86.13	44.86	88.88	46.26	87.40	46.40	86.29	73.78	
	Using $\mathbb{P}_{\text{in}}$ and $\mathbb{P}_{\text{wild}}$													
OE	2.86	99.05	40.21	88.75	4.13	99.05	1.25	99.38	22.86	94.63	14.26	96.17	73.38	
Energy (w/ OE)	2.71	99.34	34.82	90.05	3.27	99.18	2.54	99.23	30.16	94.76	14.70	96.51	72.76	
WOODS	0.17	99.80	21.87	93.73	0.48	99.61	1.24	99.54	9.95	95.97	6.74	97.73	73.91	
Medix	0.16	99.96	15.99	95.23	0.13	99.98	0.83	99.83	8.02	97.79	5.42	98.96	73.33	
(Ours)	± 0.02	± 0.00	± 0.66	± 0.14	± 0.06	± 0.02	± 0.36	± 0.06	± 0.75	± 0.30	± 0.37	± 0.10	± 0.83	

100—a 1.32-point absolute reduction over WOODS—while retaining a competitive InD accuracy of 73.33%. The modest accuracy gap is fully attributable to the experimental protocol: Medix trains on 25,000 labeled InD samples, whereas the InD-only baselines use the full 50,000-image CIFAR-100 training set. Because the two figures are not measured under matched labeled-data budgets, the gap reflects a design choice rather than any weakness of the method. The CIFAR-10 results in Table 1 tell the same story with even wider margins: Medix again surpasses all baselines, improving on WOODS by 7.21 points on PLACES365 and 5.25 points on TEXTURES in FPR95 and driving the average FPR95 to 0.80%. That the gains persist across two InD datasets, five OOD test sets, and both detection metrics is strong evidence that median-centric filtering is a broadly reliable, rather than dataset-specific, mechanism for OOD detection.

A representative visual example of Medix. To render the behavior of Algorithm 1 (Outlier Extraction) concrete, we construct a deliberately simple 2D synthetic study that visualizes how OOD samples are extracted from a wild mixture $\mathcal{S}_{\text{wild}}$ in a setting where the ground-truth labels are available. The InD data are sampled from three class-conditional multivariate Gaussians with mean vectors $[-2, 0]$, $[2, 0]$, and $[0, 2\sqrt{3}]$ and shared covariance $0.25 \cdot I$, while the OOD data are drawn

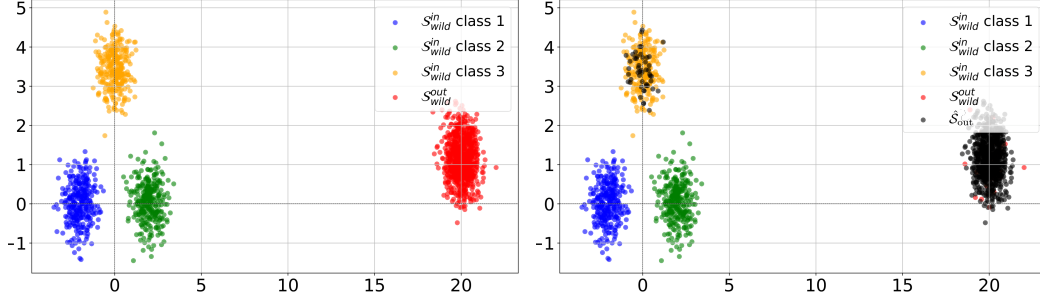


Figure 2: Medix applied to a 2D synthetic wild mixture with known ground truth. (a) The InD data $\mathcal{S}_{\text{wild}}^{\text{in}}$ (inliers drawn from three multivariate Gaussians) alongside the OOD data $\mathcal{S}_{\text{wild}}^{\text{out}}$. (b) The outliers $\hat{\mathcal{S}}_{\text{out}}$ flagged by Medix (black) align closely with the ground-truth OOD points: only 12.5% of $\hat{\mathcal{S}}_{\text{out}}$ are misclassified InD samples, instantiating in a controlled setting the low-error behavior predicted by Theorem 4.1.

353 from a well-separated Gaussian $\mathcal{N}([20, 2\sqrt{3}], 0.25 \cdot I)$. As Figure 2 shows, Medix recovers the
 354 outliers with high fidelity, flagging 87.5% of the true OOD samples—equivalently, only 12.5% of the
 355 extracted set are misclassified inliers. This controlled study, in which ground truth is known, provides
 356 a direct empirical instance of the low-error behavior characterized by our analysis, complementing
 357 the benchmark results where such labels are unavailable.

358 **Additional studies and insights.** Space constraints lead us to defer a substantial body of supporting
 359 experiments to Appendix A, each isolating a distinct facet of Medix’s behavior and, taken together,
 360 addressing the practical questions most likely to bear on deployment: (1) hyperparameter ablations
 361 and sensitivity analysis (Appendix A.2), establishing that detection quality degrades gracefully rather
 362 than sharply as ϵ and k vary; (2) a comparison between the element-wise and geometric median
 363 (Appendix A.1), showing that the EWM is more sensitive to distributional shift and therefore the more
 364 effective filter; (3) a comparison against methods with competitive contrastive learning objectives
 365 (Appendix A.3), where Medix again outperforms by a wide margin; (4) an evaluation on large-scale
 366 data under the demanding unseen-OOD setting $\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$ (Appendix A.4), in which the wild and
 367 test OOD distributions are deliberately disjoint and Medix retains a clear lead; (5) a profiling of
 368 computation and memory cost (Appendix A.6), confirming feasibility at the tested scale; (6) an
 369 analysis of pseudo-label quality (Appendix A.5), showing resilience to noisy and low-confidence
 370 pseudo-labels; and (7) a comparison with semi-supervised open-set recognition methods (Appendix
 371 A.7).

372 6 Related Work

373 OOD detection has attracted intense and sustained interest in recent years (Fort et al., 2021; Yang
 374 et al., 2024; Fang et al., 2022; Zhu et al., 2022; Yang et al., 2022; Wang et al., 2022c; Galil et al.,
 375 2023; Djurisic et al., 2023; Tao et al., 2023; Zheng et al., 2023; Wang et al., 2022b, 2023b; Uppaal
 376 et al., 2023; Zhu et al., 2023; Bai et al., 2023; Ming and Li, 2024; Zhang et al., 2023; Ghosal et al.,
 377 2024; Abbas et al., 2025), and the resulting literature organizes naturally along two dominant axes:
 378 post hoc scoring and training-time regularization. The first strategy designs scoring functions that
 379 gauge how far an input departs from the InD distribution, spanning distance-based methods (Lee
 380 et al., 2018b; Tack et al., 2020b; Ren et al., 2021; Sehwal et al., 2021; Sun et al., 2022; Du et al.,
 381 2022a; Ming et al., 2023; Ren et al., 2022), gradient-based scores (Huang et al., 2021b), energy-based
 382 scores (Liu et al., 2020c; Wang et al., 2021; Wu et al., 2023), confidence-based criteria (Bendale
 383 and Boult, 2016; Hendrycks and Gimpel, 2017; Liang et al., 2018), and Bayesian methods (Gal
 384 and Ghahramani, 2016; Lakshminarayanan et al., 2017; Maddox et al., 2019; Malinin and Gales,
 385 2019; Wen et al., 2020; Kristiadi et al., 2020). Medix shares the gradient-space viewpoint of the
 386 gradient-based scores but differs in purpose: rather than scoring individual test inputs post hoc, it
 387 aggregates gradients through a robust median statistic to mine outliers from an unlabeled training
 388 mixture.

A second family of approaches instead regularizes the model during training (Malinin and Gales, 2018; Geifman and El-Yaniv, 2019; Hein et al., 2019; Meinke and Hein, 2020; Jeong and Kim, 2020; Liu et al., 2020a; Van Amersfoort et al., 2020; Yang et al., 2021; Wei et al., 2022; Du et al., 2022b, 2023; Wang et al., 2023a), typically by driving the model toward lower confidence (Lee et al., 2017; Hendrycks et al., 2019) or higher energy (Liu et al., 2020c; Du et al., 2022c; Ming et al., 2022) on OOD data. The defining limitation of this line of work is its reliance on a clean auxiliary OOD dataset—precisely the assumption that Medix dispenses with by operating on unlabeled mixed data.

Closest to our setting, several studies (Zhou et al., 2021; Katz-Samuels et al., 2022b; He et al., 2023) relax the labeled-only assumption by incorporating unlabeled wild data (Katz-Samuels et al., 2022a; Geng et al., 2025), yet stop short of a principled outlier-mining mechanism. Du et al. (2024a,b) do introduce explicit outlier filtering, but their thresholding rule differs fundamentally from our median-centric formulation: where they threshold a scalar score, Medix minimizes the deviation of a robust coordinate-wise median statistic and inherits the median’s classical breakdown properties. A further and consequential distinction is structural. Katz-Samuels et al. (2022a); Du et al. (2024a) presume batch-level mixing, in which each batch contains a fixed InD/OOD ratio—an assumption that breaks down for large outsourced datasets whose samples are mixed randomly rather than by batch. Medix removes this assumption entirely, operating directly at the dataset level without any batch-level structure, and thereby applies to a strictly broader and more realistic class of deployment scenarios. Our filtering stage is also naturally read as robust data pruning repurposed for detection—identifying and separating contaminated samples rather than merely discarding them—and as a form of outlier mining that yields, rather than assumes, the auxiliary OOD set consumed by outlier-exposure training. Finally, a related body of work employs positive-unlabeled learning to train classifiers from positive and/or unlabeled data (Letouzey et al., 2000; Hsieh et al., 2015; Plessis et al., 2015; Niu et al., 2016; Gong et al., 2018; Chapel et al., 2020; Garg et al., 2021; Xu and Denil, 2021; Garg et al., 2022; Zhao et al., 2022; Acharya et al., 2022); these methods aim only to distinguish $\mathbb{P}_{\text{out}}$ from $\mathbb{P}_{\text{in}}$ and do not concurrently train an OOD detector, whereas Medix couples median-centric outlier identification to the training of a dedicated detector and situates the identification step within an explicit two-sided error analysis.

7 Conclusions

We introduced Medix, a median-centric framework for OOD detection that turns unlabeled in-the-wild data into a reliable training signal. By exploiting the classical robustness of the median, Medix filters outliers from mixed unlabeled data at the dataset level and thereby enables the training of a dependable OOD detector. On the theoretical side, our analysis of the idealized median-filtering objective delivers a two-sided characterization: bounds on the inlier misclassification rate show that median filtering remains robust even under substantial OOD contamination (up to 50%), with error governed by sub-Gaussian concentration and contamination effects, while complementary bounds control the OOD misclassification rate whenever the two gradient distributions are adequately separated and OOD points are not a small minority. Empirically, the picture is unambiguous. Across two InD datasets, five OOD test sets, and 20 baselines, Medix attains the best FPR95 and AUROC in every reported comparison—reducing average FPR95 by 40.98 points relative to the strong InD-only baseline KNN+, surpassing state-of-the-art wild-data and contrastive methods including WOODS and DRL, and extracting outliers at an error rate as low as 12.5% on synthetic data with known ground truth. Together, the analysis and the experiments establish median-centric filtering as a principled and practical mechanism for OOD detection in open-world deployment, and point toward extensions—coupling with generative outlier synthesis and adaptation to drifting OOD distributions—as natural next steps.

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695 A Additional Studies

696 A.1 Element-wise Median vs Geometric Median

697 A natural question is why Medix adopts
 698 than the geometric median (Acharya et al.,
 699 To settle the question empirically, we conducted a controlled experiment with CIFAR-10 as the InD dataset and SVHN as the OOD dataset. Here $\mathcal{S}_{\text{wild}}$ is initialized with 5k CIFAR-10 samples held disjoint from the data used to train $\phi_{\mathcal{S}_{\text{in}}}$ and to compute $\bar{\nabla}_{\text{in}}$. We then inject SVHN OOD samples incrementally and track two quantities: the L_2 -norm deviation between $\bar{\nabla}_{\text{in}}$ and the element-wise median of the wild gradients, and the proportion of injected OOD samples flagged for removal. As Figure 3 shows, at every fixed number of injected OOD samples the element-wise median flags a substantially higher proportion of them as outliers than the geometric median does. The element-wise median is thus markedly more responsive to distributional shift—a direct consequence of its coordinate-wise construction—which establishes it as the more effective and reliable filtering statistic for Medix.

715 A.2 Hyperparameter 716 Selection and Sensitivity Analysis

717 We ablate the sensitivity of Medix to its filtering hyperparameters ϵ and k , using CIFAR-100 as the InD dataset and SVHN as the OOD dataset. As Table 3 shows, Medix is strongly robust to variation in both parameters. With $\epsilon = 0.005$ and $k = 20000$ it attains a very low FPR95 of 0.16, and perturbing these values—reducing ϵ to 0.0005 or halving k to 10000—leaves performance competitive (FPR95 of 0.20 and 0.68, respectively), a graceful degradation rather than a sharp collapse. Across this entire range Medix continues to surpass the baselines by a wide margin. This insensitivity to precise tuning is a practical asset, since it makes Medix a dependable choice in real-world deployments where exhaustive hyperparameter search is often infeasible.

726 A.3 Comprehensive comparison with recent competitive methods

727 To stress-test Medix against the strongest available competitors, we extend the comparison to baselines built on diverse contrastive learning objectives—SSD+ (Sehwag et al., 2021), ProxyAnchor (Kim et al., 2020), and CIDER Ming et al. (2023)—which we include deliberately, despite Medix employing no contrastive objective, precisely to probe the limits of our approach against methods designed around a different inductive bias. We further add the recent CONJ (Peng et al., 2024), DRL (Zhang

the element-wise median rather
 2024) as its filtering statistic.

Table 3: Effect of the Medix filtering hyperparameters ϵ and k on FPR95 (CIFAR-100 InD, SVHN OOD). Across the tested settings, Medix remains far below all baselines, exhibiting graceful degradation rather than sharp sensitivity to the choice of hyperparameters.

Method	FPR95↓	ϵ	k
DICE	88.35	—	—
ASH	21.36	—	—
CSI	64.70	—	—
KNN+	32.21	—	—
OE	2.86	—	—
Energy	2.71	—	—
Medix	0.16	0.005	20000
Medix	0.20	0.0005	20000
Medix	0.68	0.005	10000

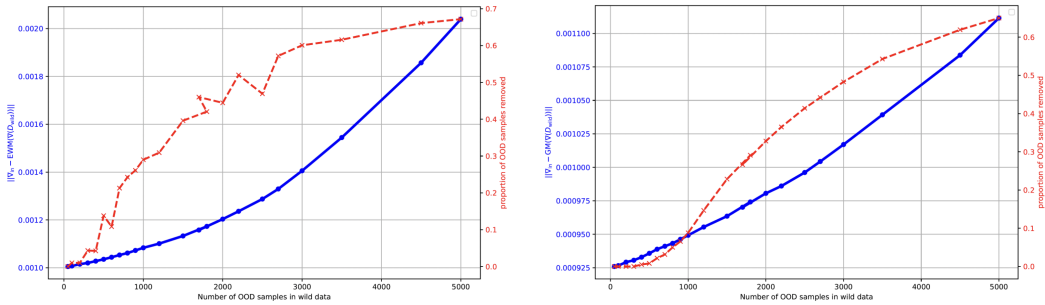


Figure 3: Element-wise median (EWM) versus geometric median (GM) as SVHN OOD samples are incrementally added to $\mathcal{S}_{\text{wild}}$ (CIFAR-10 InD, SVHN OOD): at a matched number of injected OOD samples, the EWM flags a markedly higher proportion of them as outliers, evidencing its greater sensitivity to distributional shift and motivating its selection as the filtering statistic in Medix.

Table 4: OOD detection performance of Medix versus recent competitive baselines (including contrastive-objective methods) on CIFAR-100 as InD data with a ResNet-34 backbone, across four OOD test sets and their average. Lower FPR95 and higher AUROC are better; results are averaged over five runs (with standard deviations), and the best entry in each column is in **bold**. Medix achieves the best FPR95 and AUROC on every OOD set and the best averages (8.68 FPR95, 97.76 AUROC), improving on the next-best average FPR95 (DRL, 36.43) by more than a factor of four.

Methods	OOD Datasets									
	SVHN		PLACES365		LSUN-C		TEXTURES		AVERAGE	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
MSP	78.89	79.80	84.38	74.21	83.47	75.28	86.51	72.53	83.31	75.46
Mahalanobis	87.09	80.62	84.63	73.89	84.15	79.43	61.72	84.87	79.40	79.70
ODIN	70.16	84.88	82.16	75.19	76.36	80.10	85.28	75.23	78.49	78.85
Energy	66.91	85.25	81.41	76.37	59.77	86.69	79.01	79.96	71.78	82.07
ReAct	50.93	88.75	83.55	73.10	64.02	80.31	64.40	81.95	65.72	81.03
KNN	46.25	90.39	82.08	75.44	60.85	85.61	62.39	83.95	62.89	83.85
Vim	73.42	84.62	85.34	69.34	86.96	69.74	74.56	76.23	80.07	74.98
VOS	43.24	82.80	76.85	78.63	73.61	84.69	57.57	87.31	62.82	83.36
CSI	44.53	92.65	79.08	76.27	75.58	83.78	61.61	86.47	65.20	84.79
ProxyAnchor	87.21	82.43	70.10	79.84	37.19	91.68	65.64	84.99	65.04	84.74
SSD+	31.19	94.19	77.74	79.90	79.39	85.18	66.63	86.18	63.74	86.36
KNN+	39.23	92.78	80.74	77.58	48.99	89.30	57.15	88.35	56.53	87.00
ASH	52.96	90.19	72.62	76.38	75.18	76.52	56.17	86.75	64.23	82.46
CIDER	23.09	95.16	79.63	73.43	16.16	96.33	43.87	90.42	40.69	88.84
CONJ	46.19	90.44	80.81	75.83	60.45	85.90	62.13	83.77	62.40	83.99
DRL	20.15	94.07	76.64	77.55	16.97	94.63	31.97	92.09	36.43	89.59
Medix	0.48	99.87	24.52	93.42	0.73	99.84	8.99	97.92	8.68	97.76
(Ours)	± 0.05	± 0.02	± 1.76	± 0.36	± 0.08	± 0.03	± 0.72	± 0.24	± 0.65	± 0.16

et al., 2024), Vim (Wang et al., 2022a), and VOS (Du et al., 2022c). For this experiment we use CIFAR-100 as the InD dataset and train a ResNet-34 model¹ following Ming et al. (2023); Zhang et al. (2024): stochastic gradient descent with momentum 0.9 and weight decay 10^{-4} for 500 epochs, initial learning rate 0.5 with cosine scheduling, and batch size 512, using the checkpoints of Ming et al. (2023)². As Table 4 shows, Medix outperforms every one of these competitive baselines by a substantial margin—including the contrastive-objective methods—attaining an average FPR95 of 8.68% and average AUROC of 97.76%, against a next-best average FPR95 of 36.43% (DRL). Beyond this margin in raw detection accuracy, Medix is distinguished by the accompanying two-sided error analysis, a theoretical footing that the great majority of these alternatives do not provide.

A.4 Evaluation on Large-Scale, Complex Unseen OOD Settings

We now examine whether Medix scales to large wild OOD data under the more demanding unseen-OOD regime, which directly probes the concern that the wild and test OOD distributions might otherwise be too closely related. This setting is harder on two counts: (1) it employs a large-scale wild OOD dataset ($\mathbb{P}_{\text{out}}$), which raises both data complexity and computational cost; and (2) the test-time OOD distribution ($\mathbb{P}_{\text{out}}^{\text{test}}$) is deliberately chosen to be disjoint from the wild OOD data seen in training, i.e., $\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$, mirroring the distribution shift that characterizes realistic open-world deployment. Concretely, we take CIFAR-100 as the labeled InD data, the 300K Random Images dataset Hendrycks et al. (2019) as the unlabeled wild OOD data, and SVHN as the test OOD data—so that the semantic gap between the large-scale, heterogeneous wild OOD data (300K Random Images) and the test OOD data (SVHN) is precisely what makes the task hard. Comparing against baselines that likewise train on the same wild data, Table 5 shows that Medix is clearly superior, delivering a substantially lower FPR (41.29 ± 1.2 , versus 68.80 to 69.81 for the wild-data baselines) and the highest AUROC (87.25 ± 0.9). This is direct evidence that

Table 5: OOD detection on large-scale unseen OOD data (CIFAR-100 InD, 300K Random Images as wild OOD, SVHN as test OOD, with disjoint distributions $\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$). Among wild-data methods trained on the identical wild set, Medix attains the lowest FPR and highest AUROC, with an FPR roughly 28 points below the next-best baseline.

Method	FPR ↓	AUROC ↑
OE	68.80 ± 2.8	82.89 ± 1.1
Energy (w/ OE)	69.81 ± 2.4	85.59 ± 1.0
WOODS	69.41 ± 2.7	86.76 ± 0.8
Medix (ours)	41.29 ± 1.2	87.25 ± 0.9

¹Since ResNet-34 is the standard backbone across these works, we adopt the same model to ensure consistency and facilitate a fair comparison.

²<https://github.com>

median-centric filtering continues to distinguish InD from OOD data even when the wild and test OOD distributions are disjoint and the wild OOD source is large and semantically diverse.

A.5 Evaluating the Impact of Pseudo-Label Quality on OOD Filtering and Robustness

A further question concerns the role of pseudo-label quality. Recall that each wild-sample gradient in (4) is evaluated at the top-1 pseudo-label $\hat{y}_{\tilde{x}_i}$ assigned by the InD classifier; it is therefore natural to ask whether the softmax confidence of these pseudo-labels influences filtering, and whether a simple pre-filtering step that discards low-confidence pseudo-labels would improve robustness. To answer this, we evaluated the effect of removing low-confidence pseudo-labels, using CIFAR-10 as the InD dataset and LSUN-Resize as the OOD dataset. We computed the softmax probability of each pseudo-labeled sample and discarded predictions below a confidence threshold of 0.6, which removed 15.98% of the samples from the training set.

Detection performance was essentially unchanged: both variants yielded an FPR95 of 0.01%, and pre-filtering nudged AUROC only marginally, from 99.98% to 99.99%. Removing low-confidence pseudo-labels thus neither helps nor hurts detection appreciably, which indicates that Medix is resilient to noisy and low-confidence pseudo-labels—the robustness of the median statistic absorbs the errors that a confidence-based pre-filter would target—and that such additional filtering steps are unlikely to yield meaningful gains.

A.6 Evaluating Computation and Memory Efficiency in Medix Filtering

We next assess whether the Medix filtering phase is computationally feasible and memory-efficient at the scale we consider. We profiled it on an NVIDIA A100-SXM4-80GB GPU using roughly 15,000 unlabeled samples per InD–OOD pair, with CIFAR-10 and CIFAR-100 as the InD datasets and LSUN-Resize as the OOD dataset. Table 6 shows that peak GPU memory stays under 100 MB and the filtering phase completes within a tractable wall-clock budget at this scale. Medix filtering is therefore computationally practical and imposes no excessive memory overhead at the tested dataset sizes; the leave-one-out median updates are the dominant cost, and scaling to substantially larger corpora is a direction we leave to future work.

Table 6: Profiling of the Medix filtering phase on an NVIDIA A100 80GB GPU ($\approx 15,000$ unlabeled samples per pair). Peak GPU memory stays under 100 MB and filtering completes within a tractable wall-clock time, confirming feasibility at the tested scale.

InD–OOD	Wall Clock Time (s)	Peak GPU Memory (MB)	Current GPU Memory (MB)
CIFAR10 – LSUN-Resize	4497.17	99.46	31.74
CIFAR100 – LSUN-Resize	5478.36	99.56	31.79

A.7 Comparison with Semi-Supervised Open-Set Recognition Methods

Medix departs from recent semi-supervised open-set recognition (SSL) methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) along three distinct axes: objective, the role assigned to outliers, and technique. First, in objective: OpenML Saito et al. (2021) and its successors seek to train a standard classifier that is robust to outliers within an SSL pipeline, whereas Medix pursues the fundamentally different goal of detecting OOD samples from unlabeled wild data and training a dedicated OOD detector. Where these SSL methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) aim to preserve classification accuracy in spite of outliers, Medix aims to improve OOD detection itself in an open-world setting devoid of labeled OOD data.

Second, in the role of outliers: SSL methods treat outliers as noise to be suppressed during training so as to protect classification performance. Medix inverts this stance, actively detecting and then exploiting the very same OOD samples as a supervisory signal for a more effective and reliable detector.

803 Third, in technique: OpenMatch rests on consistency regularization, enforcing agreement across
804 stochastic transformations of the same input through soft constraints on the classifier’s outputs that
805 encourage smooth decision boundaries. Medix instead performs median-based filtering, exploiting
806 the statistical robustness of the median—its insensitivity to extreme values—to isolate candidate
807 OOD samples, and it accompanies this procedure with error bounds governed by the contamination
808 effect (the OOD proportion) and the concentration effect (the coordinate-wise sub-Gaussian behavior
809 of InD gradients).

810 **B Broader Impacts and Limitations**

811 As machine learning systems proliferate, reliable OOD detection has become indispensable for
812 robust and trustworthy performance in real-world deployment. Medix contributes a median-centric
813 framework that strengthens OOD handling and, with it, model safety. The practical relevance
814 extends beyond the theoretical contribution: because reliable rejection of unfamiliar inputs is a
815 prerequisite for safe operation, improved OOD detection bears directly on high-stakes application
816 domains—healthcare, autonomous systems, and finance among them—and, more broadly, on the
817 trust and adoption of machine learning technologies. Promising directions for future work include
818 coupling Medix with generative models for outlier synthesis and adapting it to dynamic environments
819 in which the OOD distribution drifts over time.

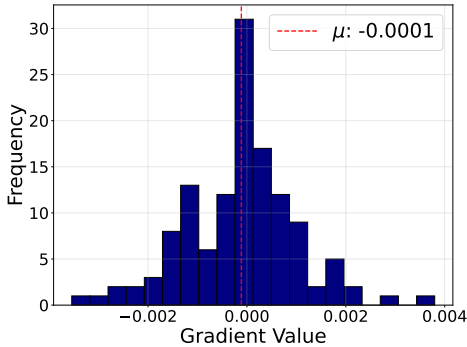
820 Medix’s principal limitation is its reliance on a mixture of unlabeled InD and OOD data, which in
821 practice may be noisy, corrupted, or inconsistently sampled. This assumption is neither unique to our
822 method nor arbitrary: it reflects the inherent uncertainty of real-world deployment and is a standard
823 premise in robust learning theory. Two further boundaries of the present work merit explicit statement.
824 First, our theoretical guarantees concern the idealized median-filtering objective rather than the
825 greedy solver of Algorithm 1, and the outlier bound is informative only when OOD samples are not a
826 small minority (π not too far below $1/2$); establishing an approximation guarantee for the heuristic
827 and tightening the analysis in the low- π regime are natural next steps. Second, our computational
828 profiling establishes feasibility at the tested scale, and confirming it on substantially larger datasets and
829 models remains to be done. That said, we take deliberate care to make our assumptions explicit and
830 verifiable—formalizing them in the main theorems, stating a precise geometric separation condition
831 in Theorem 4.2, and empirically checking the sub-Gaussianity of gradients in Remark 4.3—and we
832 further relax the tail assumption in Theorem C.3, which dispenses with sub-Gaussian tails and shows
833 that the framework remains resilient under merely bounded fourth moments. These results indicate
834 that Medix’s simplifying assumptions are both interpretable and relaxable, providing a principled
835 foundation for further advances in OOD detection under realistic noise conditions.

836 C Proof of Theorem

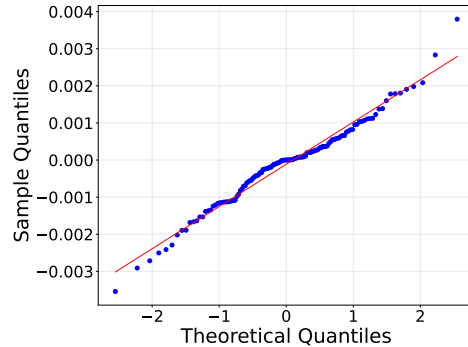
837 For clarity and consistency across the theoretical and empirical sections, Table 7 collects the key
 838 notation used throughout the paper—gradient-related quantities, sample sizes, filtering terms, and the
 839 parameters entering our concentration bounds. Unless stated otherwise, boldface denotes vectors, $\mathcal{X}$
 840 denotes the input space, and every gradient is computed with respect to a fixed pretrained model f_ϕ .

Table 7: Notation summary. Boldface denotes vectors and $\mathcal{X}$ denotes the input space.

Symbol	Meaning
$\mathcal{X}$	Input space (e.g., images in $[0, 1]^d$)
f_ϕ	Model parameterized by ϕ
$\ell(f_\phi(x))$	Loss function evaluated at input x
$\nabla \ell(f_\phi(x))$	Gradient of loss at point x
$\mathbf{g}(x)$	Shorthand for $\nabla \ell(f_\phi(x))$, gradient vector
$\bar{\mathbf{g}}_{\text{in}}$	Empirical mean of gradients over InD points
m	Total number of unlabeled samples in $\mathcal{S}_{\text{wild}}$
$m_{\text{in}} = (1 - \pi)m$	Number of InD samples
$m_{\text{out}} = \pi m$	Number of OOD samples
$\pi \in (0, 1)$	OOD contamination ratio
$\mathcal{S}_{\text{wild}}$	Unlabeled mixture of InD and OOD data
$\mathcal{S}_{\text{in}}^*$	Inliers selected by the median filter
$\mathcal{S}_{\text{out}}^*$	OOD points selected by the median filter
$\mathcal{B}_\varepsilon(\bar{\mathbf{g}}_{\text{in}})$	Ball of radius ε centered at $\bar{\mathbf{g}}_{\text{in}}$
m_ε	Number of InD points inside $\mathcal{B}_\varepsilon$
m_ε^*	Number of selected inliers inside $\mathcal{B}_\varepsilon$
m_{sep}^*	Number of well-separated OODs in selected set
ERR_{in}	Fraction of InD points misclassified as OOD
ERR_{out}	Fraction of OOD points misclassified as InD
σ^2	Sub-Gaussian proxy variance of each gradient coordinate
ε	Threshold for gradient deviation, $\varepsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$
$\delta \in (0, 1)$	Confidence parameter for concentration bounds
d	Dimensionality of gradient vectors



(a) InD gradient histogram.



(b) InD gradient Q-Q plot.

Figure 4: Empirical evidence that InD sample gradients are coordinate-wise sub-Gaussian on the datasets we study. (left) The histogram of gradient values (CIFAR-100 InD data) concentrates about the mean with light tails, consistent with sub-Gaussianity. (right) The Q-Q plot of empirical InD gradient quantiles against theoretical Gaussian quantiles tracks the reference line closely, corroborating the sub-Gaussian assumption that underlies Theorem 4.1.

841 C.1 Bound on the Inlier Misclassification Rate

842 **Theorem C.1** (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are*
 843 *i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\epsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$, and*
 844 *fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification*
 845 *rate of the element-wise median (EWM) filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

846 *Proof.* The argument proceeds in two steps. **Step 1** establishes a high-probability concentration
 847 bound for the gradients of InD samples around their mean $\bar{\nabla}_{\text{in}}$ and designates any InD sample
 848 whose gradient lies within this high-probability region a *good inlier*. **Step 2** applies a “swapping”
 849 argument to the EWM objective, showing that excluding more than a $(2\eta + \frac{\pi}{2}(1-\pi))$ fraction of *good*
 850 *inliers*—where η denotes the fraction of InD samples whose gradients fall outside the high-probability
 851 region—would strictly decrease the objective and hence contradict the optimality of the minimizing
 852 subset. Combining the two steps yields the stated upper bound on the inlier misclassification rate
 853 ERR_{in} .

854 **Step 1: Concentration of InD Gradients.** Let $\nabla \ell(f_\phi(\mathbf{x})) \in \mathbb{R}^d$ be the gradient (with respect to ϕ)
 855 of a loss evaluated at an InD sample $\mathbf{x}$. Suppose each coordinate of this gradient is sub-Gaussian with
 856 variance proxy σ^2 , centered at the same mean vector $\bar{\nabla}_{\text{in}}$. That is, for each coordinate $j \in \{1, \dots, d\}$,

$$\mathbb{P}\left(|\nabla \ell(f_\phi(\mathbf{x}))_j - \bar{\nabla}_{\text{in},j}| \geq \epsilon\right) \leq 2 \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (6)$$

857 Define

$$\mathbf{v} := \nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}} \in \mathbb{R}^d.$$

858 It is straightforward to show that

$$\|\mathbf{v}\|_2 \leq \sqrt{d} \|\mathbf{v}\|_\infty,$$

859 so if $\|\mathbf{v}\|_2 > \epsilon \sqrt{d}$ then there must exist at least one coordinate j with $|v_j| > \epsilon$.

860 Hence, combining (6) across $j = 1, \dots, d$ via a union bound yields

$$\mathbb{P}\left(\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (7)$$

861 Let $m_{\text{in}} = (1 - \pi)m$ denote the total number of InD points in the set $\mathcal{S}_{\text{wild}}$. We define an InD point
 862 $\mathbf{x}$ to be *bad* if

$$\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}. \quad (8)$$

863 Our goal is to show that, with high probability, only a small fraction of these m_{in} inliers can be bad.

864 From our sub-Gaussian assumption and applying the union bound across the d coordinates, we define
 865 a constant $p \in [0, 1]$ as follows:

$$p := \mathbb{P}\left(\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (9)$$

866 Here, $\mathbf{x}$ is a generic InD sample, and we assume that the samples are i.i.d. across the m_{in} inliers.

867 Let X be the random variable denoting the number of bad inliers among the m_{in} samples. For each
 868 inlier $\mathbf{x}_i$, define the indicator variable

$$I_i := \begin{cases} 1, & \text{if } \|\nabla \ell(f_\phi(\mathbf{x}_i)) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

869 Then, X can be expressed as

$$X = \sum_{i=1}^{m_{\text{in}}} I_i. \quad (11)$$

870 By the linearity of expectation, we have

$$\mathbb{E}[X] = \sum_{i=1}^{m_{\text{in}}} \mathbb{E}[I_i] \quad (12)$$

$$= \sum_{i=1}^{m_{\text{in}}} \mathbb{P}(\mathbf{x}_i \text{ is bad}) \quad (13)$$

$$\stackrel{\text{from (9)}}{\leq} m_{\text{in}} \cdot p \quad (14)$$

$$\leq m_{\text{in}} \cdot 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (15)$$

871 Observing that X is a sum of m_{in} Bernoulli(p) random variables, each bounded between 0 and 1, we
872 can apply Hoeffding's inequality to obtain, for any $t > 0$,

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (16)$$

873 Setting

$$t := \sqrt{\frac{m_{\text{in}} \log(\frac{1}{\delta})}{2}}, \quad (17)$$

874 we obtain

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (18)$$

875 With probability at least $1 - \delta$, we therefore have

$$X \leq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (19)$$

876 Dividing both sides by m_{in} yields an upper bound on the fraction of bad inliers:

$$\frac{X}{m_{\text{in}}} \leq \frac{\mathbb{E}[X]}{m_{\text{in}}} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (20)$$

$$\stackrel{\text{from (12)}}{\leq} 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (21)$$

$$= \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}. \quad (22)$$

877 Let us denote this fraction by η . By setting $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$ Thus, with probability at least
878 $1 - \delta$, no more than an $O\left(d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{m_{\text{in}}}}\right)$ fraction of inliers have gradients lying
879 outside the radius $\epsilon\sqrt{d}$ around $\bar{\nabla}_{\text{in}}$. These are precisely the *bad inliers*, while the remainder are
880 called *good inliers*.

881 **Step 2: Contradiction via a Swapping Argument** We demonstrate that if $\mathcal{S}^*$ excludes an excessive
 882 number of InD, specifically when

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (23)$$

883 then it is possible to construct a new set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than that of $\mathcal{S}^*$,
 884 thereby contradicting the optimality of $\mathcal{S}^*$.

885 Let m_ε be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 886 lies outside the closed ℓ_2 ball

$$B_\varepsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (24)$$

887 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

888 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (25)$$

889 Define the swap size as

$$m'_\varepsilon := \min \{m_\varepsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}}\}. \quad (26)$$

890 This quantity counts how many good inliers (those inside B_ε) are currently excluded from $\mathcal{S}^*$ and
 891 available for swapping in.

892 Construct a new set $\hat{\mathcal{S}}$ by removing m'_ε points from $\mathcal{S}^*$ that lie outside B_ε (OOD points or bad inliers)
 893 and inserting m'_ε good inliers from B_ε in their place. This maintains cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

894 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\varepsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (27)$$

895 which measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging, we obtain:

$$m_\varepsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (28)$$

896 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (29)$$

897 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (30)$$

898 We know that at most a η fraction of inliers are outside B_ε , and the fraction of OOD points among all
 899 samples is π . Therefore, the total number of samples in the dataset that lie outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (31)$$

900 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (32)$$

901 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (33)$$

902 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (34)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (35)$$

903 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (36)$$

904 so the points lying inside B_ε form a strict majority in $\hat{\mathcal{S}}$.

905 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of all
906 gradients in $\mathcal{S}^*$ and $\hat{X}^{(j)} = \{\hat{x}_1^{(j)}, \dots, \hat{x}_m^{(j)}\}$ the same in $\hat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

907 Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies
908 outside I_j . So for some coordinate j^* , the number of values inside I_{j^*} increases from below $m/2$ to
909 above $m/2$ due to the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\hat{X}^{(j^*)}) \in I_{j^*}, \quad (37)$$

910 which implies

$$|\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (38)$$

911 For the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unchanged or
912 increase it, so the median distance from μ_j does not increase and possibly decreases.

913 Thus, component-wise:

$$|\text{EWM}_j(G_{\hat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (39)$$

914 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (40)$$

915 This contradicts the assumption that $\mathcal{S}^*$ is a minimizer of the optimization problem, which concludes
916 the proof.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (41)$$

917 □

918 C.2 Bound on the Outlier Retention Rate

919 We now derive a non-asymptotic upper bound on the fraction of OOD points that survive in the
920 subset selected by the idealized MEDIX filter. The result parallels the inlier misclassification bound
921 (Appendix C.1), but where that argument invoked the concentration of inliers, this one instead
922 leverages a separation assumption between the OOD and InD gradient means.

923 **Theorem C.2** (Outlier Retention Bound under Vector Separation). Assume that for every OOD point
 924 $x \sim P_{\text{out}}$, the gradient $g(x) := \nabla \ell(f_{\varphi_{\text{in}}}(x)) \in \mathbb{R}^d$ satisfies:

925 **(I) Sub-Gaussianity.** Each coordinate $g_j(x)$ is sub-Gaussian with variance proxy σ_{out}^2 .

926 **(II) Vector Mean Separation.** The mean OOD gradient vector μ_{out} satisfies

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d}, \quad \text{for some } \Delta > 0.$$

927 Fix any tolerance $\varepsilon \in (0, \Delta)$ and confidence parameter $\delta \in (0, 1)$. Define:

$$p_{\text{out}}(\varepsilon) := 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right), \quad \eta_{\text{out}} := p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}.$$

928 Then with probability at least $1 - \delta$, the outlier retention rate satisfies:

$$\text{ERR}_{\text{out}} := \frac{|\mathcal{S}_{\text{in}}^* \cap \text{OOD}|}{m_{\text{out}}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1 - \pi}{2\pi}. \quad (42)$$

929 *Proof. Step 1: Concentration of OOD Gradients.*

930 By assumption, each coordinate of $g(x)$ is sub-Gaussian, and the mean vector is separated from $\bar{\nabla}_{\text{in}}$
 931 by at least $\Delta\sqrt{d}$. Therefore, for any OOD point x , the event

$$\|g(x) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \quad (43)$$

932 has probability at most $p_{\text{out}}(\varepsilon)$.

933 **Counting the “ambiguous” OOD points.** For every OOD example x_i define the indicator

$$J_i := \mathbf{1}\left\{\|g(x_i) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d}\right\}, \quad i = 1, \dots, m_{\text{out}}.$$

934 Each $J_i \sim \text{Bernoulli}(p_{\text{out}}(\varepsilon))$ and the J_i ’s are independent. Let

$$Y := \sum_{i=1}^{m_{\text{out}}} J_i, \quad \mu := \mathbb{E}[Y] = m_{\text{out}} p_{\text{out}}(\varepsilon).$$

935 Because the indicators J_i take values in $\{0, 1\}$, Hoeffding’s inequality states that for any $s > 0$

$$\Pr\{Y - \mu \geq s\} \leq \exp\left(-\frac{2s^2}{m_{\text{out}}}\right).$$

936 Choose $s = \sqrt{\frac{m_{\text{out}} \log(1/\delta)}{2}}$; then (C.2) equals δ , and with probability at least $1 - \delta$

$$Y \leq m_{\text{out}} \left(p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}\right) = m_{\text{out}} \eta_{\text{out}}.$$

937 Define the high-probability event

$$\mathcal{E}_1 := \{Y \leq m_{\text{out}} \eta_{\text{out}}\}, \quad (35)$$

938 which holds with probability at least $1 - \delta$.

939 **Step 2: Contradiction via a Swapping Argument** Suppose, toward contradiction, that

$$\text{ERR}_{\text{out}} > \eta_{\text{out}} + \frac{1 - \pi}{2\pi}. \quad (44)$$

940 Let $m_{\text{out}}^* := |\mathcal{S}_{\text{in}}^* \cap \text{OOD}|$, and define $m_{\varepsilon}^* := \#\{\text{ambiguous OODs in } \mathcal{S}_{\text{in}}^*\}$. Then, under $\mathcal{E}_1$,

$$m_{\text{sep}}^* := m_{\text{out}}^* - m_\varepsilon^* > \frac{1-\pi}{2\pi} m_{\text{out}} = \frac{1}{2} m_{\text{in}}. \quad (45)$$

Thus, more than half the points in $\mathcal{S}_{\text{in}}^*$ are separated OODs whose gradients lie outside the ball

$$B_\varepsilon := \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (46)$$

By the definition of the coordinate-wise median, this implies that in every coordinate, the majority of entries in $G_{\mathcal{S}_{\text{in}}^*}^{(j)}$ lie outside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, hence:

$$\|\text{EWM}(G_{\mathcal{S}_{\text{in}}^*}) - \bar{\nabla}_{\text{in}}\|_2^2 \geq \varepsilon^2 d. \quad (47)$$

Now consider a new subset $\hat{\mathcal{S}}$ formed by swapping the m_{sep}^* separated OODs in $\mathcal{S}_{\text{in}}^*$ with an equal number of InD points whose gradients lie inside B_ε . Such inliers exist because fewer than $m_{\text{out}}\eta_{\text{out}}$ are ambiguous, and the total number of InD points is m_{in} .

Then at least half the entries in every coordinate of $G_{\hat{\mathcal{S}}}^{(j)}$ lie inside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, implying:

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2^2 \leq \varepsilon^2 d. \quad (48)$$

Combining with (47) yields a contradiction:

$$\|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\hat{\mathcal{S}}})\|_2^2 < \|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}_{\text{in}}^*})\|_2^2. \quad (49)$$

This contradicts the optimality of $\mathcal{S}_{\text{in}}^*$, so the assumption must be false. We therefore conclude that, with probability at least $1 - \delta$, the outlier retention rate is bounded by

$$\text{ERR}_{\text{out}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1-\pi}{2\pi}, \quad (50)$$

□

C.3 Theorem C.3 without Sub-Gaussianity

Theorem C.3 (Inlier Misclassification Bound without Sub-Gaussianity). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and that each coordinate has variance σ^2 and finite fourth moment bounded by μ_4 , i.e.,*

$$\mathbb{E}[(\nabla \ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^2 = \sigma^2, \quad \mathbb{E}[(\nabla \ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^4 \leq \mu_4.$$

Let $m_{\text{in}} = (1 - \pi)m$ denote the number of InD samples in $\mathcal{S}_{\text{wild}}$, and fix any tolerance $\varepsilon > \sigma$ and confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the element-wise median (EWM) filtering rule satisfies:

$$\text{ERR}_{\text{in}} \leq \underbrace{2 \left(\frac{\mu_4 - \sigma^4}{d(\varepsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \right)}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

960 **Differences with the original bound.** The two bounds differ precisely in how their dimensional
 961 decay depends on the tail assumption. The sub-Gaussian bound carries the decay term

$$2 \exp \left(-\frac{\epsilon^2 d}{2\sigma_{\text{sub}}^2} \right),$$

962 which decreases exponentially in the dimension d and is therefore very tight for large d . Relaxing the
 963 assumption to finite fourth moments replaces this term with

$$\frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2},$$

964 which decays only as $O(1/d)$ and yields a correspondingly looser bound. The trade-off is thus made
 965 explicit: the original result demands sub-Gaussian tails, whereas the relaxed result tolerates heavier
 966 tails at the cost of a weaker dimensional rate. Crucially, both bounds retain the same finite-sample
 967 term

$$\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}},$$

968 which shrinks with m_{in} irrespective of the tail assumption—so the qualitative robustness of median
 969 filtering is preserved in either regime, and only the dimensional rate is affected.

970 *Proof.* The proof mirrors the structure of Theorem 4.1, comprising two steps: (1) a concentra-
 971 tion bound on the fraction of “bad” InD samples—those far from ∇_{in} —now derived without sub-
 972 Gaussianity, and (2) the identical swapping argument used to bound the inlier misclassification rate.
 973 The sole substantive change is that the sub-Gaussian tail bound is replaced by a moment-based bound
 974 obtained from the finite fourth-moment assumption via Chebyshev’s inequality.

975 **Step 1: Concentration of InD Gradients** Define the gradient deviation for an InD sample x as:

$$v = \nabla \ell(f_\phi(x)) - \nabla_{\text{in}} \in \mathbb{R}^d, \quad (51)$$

976 where $\nabla_{\text{in}} = \frac{1}{n} \sum_{(x_i, y_i) \in S_{\text{in}}} \nabla \ell(f_\phi(x_i))$ is the mean gradient over the labeled InD training set S_{in} ,
 977 and we assume ∇_{in} is the population mean. An InD sample is classified as “bad” if:

$$\|v\|_2 > \epsilon\sqrt{d}, \quad (52)$$

978 and “good” otherwise. Our goal is to bound the fraction of bad inliers among the m_{in} InD samples in
 979 S_{wild} .

980 Since $\|v\|_2^2 = \sum_{j=1}^d v_j^2$, we analyze the ℓ_2 norm via the sum of squared coordinates. Assume that
 981 the coordinates $v_j = \nabla \ell(f_\phi(x))_j - \nabla_{\text{in},j}$ are independent across $j = 1, \dots, d$, each with:

$$\mathbb{E}[v_j] = 0 \quad (\text{assuming } \nabla_{\text{in},j} = \mathbb{E}[\nabla \ell(f_\phi(x))_j]), \quad (53)$$

$$\mathbb{E}[v_j^2] = \sigma^2, \quad \mathbb{E}[v_j^4] \leq \mu_4. \quad (54)$$

982 Thus:

$$\mathbb{E}[\|v\|_2^2] = \mathbb{E} \left[\sum_{j=1}^d v_j^2 \right] = d\sigma^2, \quad (55)$$

$$\text{Var}(\|v\|_2^2) = \text{Var} \left(\sum_{j=1}^d v_j^2 \right) = \sum_{j=1}^d \text{Var}(v_j^2). \quad (56)$$

983 Compute the variance of v_j^2 :

$$\text{Var}(v_j^2) = \mathbb{E}[v_j^4] - (\mathbb{E}[v_j^2])^2 \leq \mu_4 - \sigma^4, \quad (57)$$

984 so:

$$\text{Var}(\|v\|_2^2) \leq d(\mu_4 - \sigma^4). \quad (58)$$

985 Using Chebyshev's inequality for $\|v\|_2^2$:

$$\mathbb{P}(|\|v\|_2^2 - d\sigma^2| \geq t) \leq \frac{d(\mu_4 - \sigma^4)}{t^2}. \quad (59)$$

986 We need $\mathbb{P}(\|v\|_2 > \epsilon\sqrt{d})$, which corresponds to $\|v\|_2^2 > \epsilon^2 d$. Set the threshold:

$$\|v\|_2^2 > \epsilon^2 d \implies \|v\|_2^2 - d\sigma^2 > \epsilon^2 d - d\sigma^2 = d(\epsilon^2 - \sigma^2). \quad (60)$$

987 Thus:

$$\mathbb{P}(\|v\|_2^2 > \epsilon^2 d) = \mathbb{P}(\|v\|_2^2 - d\sigma^2 > d(\epsilon^2 - \sigma^2)) \leq \frac{d(\mu_4 - \sigma^4)}{[d(\epsilon^2 - \sigma^2)]^2} = \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}, \quad (61)$$

988 provided $\epsilon > \sigma$ so that $\epsilon^2 - \sigma^2 > 0$. Define:

$$p = \mathbb{P}(\|\nabla \ell(f_\phi(x)) - \nabla_{\text{in}}\|_2 > \epsilon\sqrt{d}) \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (62)$$

989 Let X be the number of bad inliers among the m_{in} InD samples in S_{wild} , where each InD sample is
990 bad with probability at most p . Then:

$$\mathbb{E}[X] \leq m_{\text{in}}p \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (63)$$

991 Since $X = \sum_{i=1}^{m_{\text{in}}} I_i$, where $I_i = 1$ if the i -th InD sample is bad and 0 otherwise, and I_i are i.i.d.
992 Bernoulli(p) variables bounded in $[0, 1]$, apply Hoeffding's inequality:

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (64)$$

993 Set $t = \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}$, so:

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (65)$$

994 With probability at least $1 - \delta$:

$$X \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (66)$$

995 The fraction of bad inliers is:

$$\eta = \frac{X}{m_{\text{in}}} \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}. \quad (67)$$

996 This η represents the upper bound on the fraction of InD samples whose gradients deviate from $\bar{\nabla}_{\text{in}}$
 997 by more than $\epsilon\sqrt{d}$, relying only on finite fourth moments and coordinate independence.

998 **Step 2: Contradiction via a Swapping Argument** We demonstrate that if $\mathcal{S}^*$ excludes an excessive
 999 number of InD, specifically when

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (68)$$

1000 then it is possible to construct a new set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than that of $\mathcal{S}^*$,
 1001 thereby contradicting the optimality of $\mathcal{S}^*$.

1002 Let m_ϵ be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 1003 lies outside the closed ℓ_2 ball

$$B_\epsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \epsilon\sqrt{d} \right\}. \quad (69)$$

1004 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

1005 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (70)$$

1006 Define the swap size as

$$m'_\epsilon := \min \{m_\epsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}}\}. \quad (71)$$

1007 This quantity counts how many good inliers (those inside B_ϵ) are currently excluded from $\mathcal{S}^*$ and
 1008 available for swapping in.

1009 Construct a new set $\hat{\mathcal{S}}$ by removing m'_ϵ points from $\mathcal{S}^*$ that lie outside B_ϵ (OOD points or bad inliers)
 1010 and inserting m'_ϵ good inliers from B_ϵ in their place. This maintains cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

1011 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\epsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (72)$$

1012 which measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging, we obtain:

$$m_\epsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (73)$$

1013 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (74)$$

1014 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (75)$$

1015 We know that at most a η fraction of inliers are outside B_ε , and the fraction of OOD points among all
 1016 samples is π . Therefore, the total number of samples in the dataset that lie outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (76)$$

1017 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (77)$$

1018 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta) m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (78)$$

1019 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (79)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (80)$$

1020 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (81)$$

1021 so the points lying inside B_ε form a strict majority in $\hat{\mathcal{S}}$.

1022 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of all
 1023 gradients in $\mathcal{S}^*$ and $\hat{X}^{(j)} = \{\hat{x}_1^{(j)}, \dots, \hat{x}_m^{(j)}\}$ the same in $\hat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

1024 Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies
 1025 outside I_j . So for some coordinate j^* , the number of values inside I_{j^*} increases from below $m/2$ to
 1026 above $m/2$ due to the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\hat{X}^{(j^*)}) \in I_{j^*}, \quad (82)$$

1027 which implies

$$|\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (83)$$

1028 For the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unchanged or
 1029 increase it, so the median distance from μ_j does not increase and possibly decreases.

1030 Thus, component-wise:

$$|\text{EWM}_j(G_{\hat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (84)$$

1031 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (85)$$

1032 This contradicts the assumption that $\mathcal{S}^*$ is a minimizer of the optimization problem, which concludes
 1033 the proof.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (86)$$

1034 □